THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the roots of $x^2 - 7x + 10 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 28 **(b)** 31 **(c)** 30 **(d)** 29

2. If the roots of $x^2 - 10x + k = 0$ differ by 2, then $k =$ 1

(a) 24 **(b)** 23 **(c)** 26 **(d)** 25

3. For which value of k does $x^2 + kx + 9 = 0$ have equal roots (take $k > 0$)? 1

(a) 8 **(b)** 7 **(c)** 6 **(d)** 5

4. The value of k for which $x = 1$ is a root of $kx^2 + 3x - 5 = 0$ with $b = 3$, $c = -5$ is: 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 3

5. The equation $kx^2 + 6x + 3 = 0$ has real roots when: 1

(a) $k < 3$ only **(b)** $k \geq 3$

(c) $k \leq 3$ and $k \neq 0$ **(d)** $k > 0$

6. If α, β are roots of $x^2 - 5x + 6 = 0$, then $\frac{1}{\alpha} + \frac{1}{\beta} =$ 1

(a) $-\frac{1}{6}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{11}{6}$ **(d)** $\frac{17}{6}$

7. If the roots of $x^2 - 9x + 14 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 54 **(b)** 55 **(c)** 52 **(d)** 53

8. If the roots of $x^2 - 12x + k = 0$ differ by 4, then $k =$ 1

(a) 31 **(b)** 33 **(c)** 34 **(d)** 32

9. For which value of k does $x^2 + kx + 16 = 0$ have equal roots (take $k > 0$)? 1

(a) 7 **(b)** 10 **(c)** 9 **(d)** 8

10. The value of k for which $x = 2$ is a root of $kx^2 + -3x - 2 = 0$ with $b = -3$, $c = -2$ is: 1

(a) 2 **(b)** 4 **(c)** 1 **(d)** 3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the values of k for which $kx^2 + 4x + 1 = 0$ has real and equal roots. 2

12. If α, β are the roots of $x^2 - 6x + 5 = 0$, find $\alpha^2 + \beta^2$. 2

13. The sum of a number and its reciprocal is $\frac{5}{2}$. Find the number. 2

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14. Find the values of k for which $kx^2 + 6x + 3 = 0$ has real and equal roots. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve for x: $\frac{x+1}{x-1} + \frac{x-1}{x+1} = \frac{10}{3}$. 3

16. For what values of k does $x^2 + 5x + k = 0$ have real roots? Find the greatest integer value of k. 3

17. The sum of the squares of two consecutive odd positive integers is 74. Find them. 3

OR

Solve for x: $\frac{x+2}{x-2} + \frac{x-2}{x+2} = \frac{26}{5}$. 3

18. For what values of k does $x^2 + 7x + k = 0$ have real roots? Find the greatest integer value of k. 3

OR

The sum of the squares of two consecutive odd positive integers is 130. Find them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two taps running together fill a tank in $\frac{10}{3}$ hours. The larger tap alone takes 5 hours less than the smaller one. Find the time each takes alone. 5

OR

A motorboat goes 48 km upstream and 48 km downstream in 7 hours. If the speed of the stream is 2 km/h, find the speed of the boat in still water. 5

20. Two taps running together fill a tank in $\frac{24}{5}$ hours. The larger tap alone takes 4 hours less than the smaller one. Find the time each takes alone. 5

OR

A motorboat goes 60 km upstream and 60 km downstream in 8 hours. If the speed of the stream is 4 km/h, find the speed of the boat in still water. 5

END OF QUESTION PAPER — Total: 40 Marks

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